UNIT 5 · 9 hours

Differential and Common-Mode Operation



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Explain single-ended, differential and common-mode operation
- Define differential and common-mode gain
- Calculate CMRR in ratio and dB
- Explain the concept of virtual ground

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

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Modes of operation

Single-ended

One input is grounded (v1 = 0 or v2 = 0). Output = A × other input.

Differential mode

Inputs equal and opposite (v1 = −v2). Output = Ad (v2 − v1).

Common mode

Inputs identical (v1 = v2 = vcm). Ideal output = 0; practical output = Acm vcm.

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Output equation

$v_o = A_d v_d + A_{cm} v_{cm}$

general

$v_d = v_2 - v_1$

difference

$v_{cm} = (v_1 + v_2)/2$

average

- Ideal op-amp: $A_{cm} = 0$, so v_o depends only on v_d.
- Practical: small $A_{cm} \neq 0 \rightarrow$ noise common to both inputs leaks to the output.

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CMRR

CMRR = Ad / Acm

ratio

CMRR (dB) = 20 log10 (Ad / Acm)

decibels

- High CMRR means good rejection of common noise: hum, interference picked up equally on both leads.
- 741: 70–90 dB typical.
- Essential in instrumentation amplifiers and sensor circuits.

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Numerical: CMRR and output

Problem An op-amp has Ad = 10^5 and Acm = 1. Find CMRR in dB. For v1 = 100.0 mV and v2 = 100.1 mV find the output.

- CMRR = $10^5/1 = 10^5 \rightarrow 20 \log10(10^5) = 100$ dB
- vd = 0.1 mV; vcm = 100.05 mV
- vo = $10^5 \times 0.1$ mV + 1×100.05 mV = 10 V + 0.1 V

Answer: CMRR = 100 dB; vo ≈ 10.1 V (0.1 V error from common mode)

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Virtual ground

- Output is limited by the supply, e.g. $|vo| \leq 13$ V, while $A \approx 10^5$.
- So the input difference vd = vo/A is tiny (e.g. 5 V/5000 = 1 mV in closed loop).
- Virtual short: v1 ≈ v2. If the non-inverting input is grounded, the inverting input is at 0 V: virtual ground.
- No current flows into the op-amp terminals: it is a virtual short, not a real one.

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Key takeaways

- vo = Ad vd + Acm vcm
- CMRR = Ad/Acm; in dB = 20 log10
- High CMRR rejects noise common to both inputs
- Virtual ground: v1 ≈ v2 with no input current

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Practice questions

Q1. Explain the three modes of op-amp operation.

Q2. Ad = 2000, Acm = 0.2. Find CMRR in dB.

Q3. What is virtual ground and why does it occur?

Q4. Why is high CMRR important for sensor signals?